

Working Notes for $D_{s1} \rightarrow D_s \gamma$ in Light-Cone QCD Sum Rules

Ds gamma project

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Abstract

These notes are a living derivation for the light-cone QCD sum-rule calculation of $D_{s1}(2460) \rightarrow D_s \gamma$ and $D_{s1}(2536) \rightarrow D_s \gamma$. The purpose is pedagogical as well as practical: each step is written so that a master's student can follow the choices, conventions, and algebra before the final paper-level formulas are assembled.

1 Conventions

We use the mostly-minus metric

$$g_{\mu\nu} = \text{diag}(+1, -1, -1, -1), \quad (1)$$

and

$$\{\gamma_\mu, \gamma_\nu\} = 2g_{\mu\nu}, \quad \sigma_{\mu\nu} = \frac{i}{2}[\gamma_\mu, \gamma_\nu], \quad \gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3. \quad (2)$$

The Levi-Civita convention is

$$\epsilon^{0123} = +1, \quad \epsilon_{0123} = -1, \quad (3)$$

so that

$$\text{Tr}[\gamma_5 \gamma_\mu \gamma_\nu \gamma_\alpha \gamma_\beta] = -4i \epsilon_{\mu\nu\alpha\beta}. \quad (4)$$

Our Fourier convention for a propagator is

$$S(x) = \int \frac{d^4k}{(2\pi)^4} e^{-ik \cdot x} \tilde{S}(k). \quad (5)$$

2 Currents and Momentum Assignment

The external momenta are

$$P = p + q, \quad (6)$$

where P is the initial axial-meson momentum, p is the final pseudoscalar D_s momentum, and q is the real-photon momentum.

The pseudoscalar interpolating current is

$$J_{D_s}(x) = \bar{s}(x) i\gamma_5 Q(x). \quad (7)$$

For the axial channel we use two independent currents,

$$J_{A\mu}^{(0)}(x) = \bar{s}(x) \gamma_\mu \gamma_5 Q(x), \quad (8)$$

$$J_{B\mu}^{(0)}(x) = \frac{i}{m_Q + m_s} \bar{s}(x) \sigma_{\mu\alpha} P^\alpha \gamma_5 Q(x). \quad (9)$$

The tensor current contracts with P^α , not p^α , because it interpolates the initial axial state. The factor $1/(m_Q + m_s)$ restores the same mass dimension as the axial-vector current.

3 Correlation Function

For a generic axial current $\Gamma_\mu \in \{\gamma_\mu \gamma_5, i\sigma_{\mu\alpha} P^\alpha \gamma_5 / (m_Q + m_s)\}$ we study

$$\Pi_\mu(p, q) = i \int d^4x e^{ip \cdot x} \langle \gamma(q, \varepsilon) | T \{ \bar{s}(x) \Gamma_\mu Q(x) \bar{Q}(0) i\gamma_5 s(0) \} | 0 \rangle. \quad (10)$$

The target Lorentz structure for the $1^+ \rightarrow 0^- \gamma$ transition is

$$(\eta \cdot \varepsilon^*)(p \cdot q) - (\eta \cdot q)(p \cdot \varepsilon^*). \quad (11)$$

On the correlator side we therefore project onto an equivalent transverse structure built from the open axial index and the photon polarization.

4 Wick Contraction and OPE Bookkeeping

The OPE is separated into a hard-photon sector and a soft-photon sector:

$$\Pi_\mu^{\text{OPE}} = \Pi_\mu^{\text{hard}} + \Pi_\mu^{\text{soft}}. \quad (12)$$

This split is important because the two terms describe different physics and must not be double counted.

4.1 Hard photon emission

In the hard sector the photon is emitted perturbatively from either the heavy quark or the strange quark. After Wick contraction one obtains

$$\Pi_\mu^{\text{hard}} = iN_c \int d^4x e^{ip \cdot x} \text{Tr}_D \left[\Gamma_\mu S_Q^\gamma(x) i\gamma_5 S_s^0(-x) + \Gamma_\mu S_Q^0(x) i\gamma_5 S_s^\gamma(-x) \right], \quad (13)$$

where $N_c = 3$ and Tr_D denotes the Dirac trace. The free massive propagator is

$$S_q^0(x) = \int \frac{d^4k}{(2\pi)^4} e^{-ik \cdot x} \frac{i(\not{k} + m_q)}{k^2 - m_q^2}. \quad (14)$$

We keep explicit powers of m_s , including possible m_s^2 terms.

The first-order electromagnetic background-field correction for a generic quark flavor q is

$$S_q^\gamma(x) = -ie_q \int \frac{d^4k}{(2\pi)^4} e^{-ik \cdot x} \int_0^1 du \left[\frac{\not{k} + m_q}{2(m_q^2 - k^2)^2} F^{\alpha\beta}(ux) \sigma_{\alpha\beta} + \frac{ux_\alpha}{m_q^2 - k^2} F^{\alpha\beta}(ux) \gamma_\beta \right]. \quad (15)$$

In Eq. (13) this generic formula is used twice:

$$S_Q^\gamma(x) = S_q^\gamma(x) \big|_{q \rightarrow Q, m_q \rightarrow m_Q, e_q \rightarrow e_Q}, \quad (16)$$

$$S_s^\gamma(x) = S_q^\gamma(x) \big|_{q \rightarrow s, m_q \rightarrow m_s, e_q \rightarrow e_s}. \quad (17)$$

Thus the hard sector contains both heavy-line and strange-line electromagnetic background-field insertions. For a real photon we write

$$F^{\alpha\beta}(ux) = i(\varepsilon^\alpha q^\beta - \varepsilon^\beta q^\alpha) e^{iuq \cdot x}, \quad (18)$$

with the overall phase kept consistent with the Fourier convention.

Equation (15) is not a vacuum-condensate correction. It is the quark propagator in an external electromagnetic field and is included in the base analysis. What we do not include in the base hard sector are the ordinary SVZ-style condensate-expanded propagator terms such as $\langle \bar{s}s \rangle$, $\langle \bar{s}g_s \sigma G s \rangle$, or $\langle G^2 \rangle$ propagator corrections.

4.2 Equivalent momentum-space vertex form

For FeynCalc trace checks it is often simpler to use the equivalent explicit photon-vertex representation. With photon index ν ,

$$N_{Q,\mu\nu} = \text{Tr}_D[\Gamma_\mu(\not{k} + \not{q} + m_Q)\gamma_\nu(\not{k} + m_Q)\text{i}\gamma_5(\not{k} - \not{p} + m_s)], \quad (19)$$

$$N_{s,\mu\nu} = \text{Tr}_D[\Gamma_\mu(\not{k} + m_Q)\text{i}\gamma_5(\not{k} - \not{p} + m_s)\gamma_\nu(\not{k} - \not{p} + \not{q} + m_s)]. \quad (20)$$

These numerators are accompanied by the denominators

$$D_Q = [(k+q)^2 - m_Q^2][k^2 - m_Q^2][(k-p)^2 - m_s^2], \quad (21)$$

$$D_s = [k^2 - m_Q^2][(k-p)^2 - m_s^2][(k-p+q)^2 - m_s^2]. \quad (22)$$

The explicit-vertex form and the background-field form represent the same hard emission physics. We must not add both as independent contributions.

4.3 Soft photon emission: two-particle photon DAs

In the soft sector the photon is represented by its light-cone distribution amplitudes. Contracting only the heavy quark gives

$$\Pi_\mu^{\text{soft},2p} = \text{i} \int d^4x \text{e}^{\text{i}p \cdot x} [\Gamma_\mu S_Q^0(x) \text{i}\gamma_5]_{\alpha\beta} \langle \gamma(q, \varepsilon) | \bar{s}_\alpha(x) s_\beta(0) | 0 \rangle. \quad (23)$$

Fierz rearrangement converts the open spinor indices to a basis of Dirac bilinears,

$$\bar{s}_\alpha(x) s_\beta(0) = -\frac{1}{4} \sum_i (\Gamma_i)_{\beta\alpha} \bar{s}(x) \Gamma_i s(0), \quad (24)$$

so that

$$\Pi_\mu^{\text{soft},2p} = -\frac{\text{i}}{4} \int d^4x \text{e}^{\text{i}p \cdot x} \sum_i \text{Tr}_D[\Gamma_\mu S_Q^0(x) \text{i}\gamma_5 \Gamma_i] \langle \gamma(q, \varepsilon) | \bar{s}(x) \Gamma_i s(0) | 0 \rangle. \quad (25)$$

The photon matrix elements are then expressed in terms of two-particle photon DAs. Parameters such as $\chi \langle \bar{s}s \rangle$ enter here as part of the photon wavefunction normalization; they are not condensate corrections to the hard propagator.

4.4 Soft photon emission: three-particle photon DAs

To include three-particle photon DAs we must also allow one background gluon to come from the heavy-quark propagator,

$$S_Q^G(x) = -\text{i}g_s \int \frac{d^4k}{(2\pi)^4} \text{e}^{-\text{i}k \cdot x} \int_0^1 du \left[\frac{\not{k} + m_Q}{2(m_Q^2 - k^2)^2} G^{\alpha\beta}(ux) \sigma_{\alpha\beta} + \frac{ux_\alpha}{m_Q^2 - k^2} G^{\alpha\beta}(ux) \gamma_\beta \right]. \quad (26)$$

This term is not a gluon-condensate correction. It is an operator insertion that combines with photon matrix elements of the form

$$\langle \gamma(q, \varepsilon) | \bar{s}(x) \Gamma_i g_s G_{\alpha\beta}(ux) s(0) | 0 \rangle, \quad (27)$$

which define the three-particle photon DAs.

5 Staged Calculation Plan

We proceed in two stages.

Stage 1. Include the hard photon emission terms and the two-particle photon DAs:

$$\Pi_{\mu}^{\text{Stage 1}} = \Pi_{\mu}^{\text{hard}} + \Pi_{\mu}^{\text{soft},2p}. \quad (28)$$

This stage keeps all explicit m_s terms and omits ordinary condensate-expanded propagators.

Stage 2. Add the three-particle photon DA contributions generated by S_Q^G and the corresponding light-cone matrix elements:

$$\Pi_{\mu}^{\text{Stage 2}} = \Pi_{\mu}^{\text{Stage 1}} + \Pi_{\mu}^{\text{soft},3p}. \quad (29)$$

This staging keeps the derivation auditable: first we verify the core hard and two-particle DA structures, then we add the more involved background-gluon terms.

6 Current FeynCalc Artifacts

The script `scripts/step2_current_building_blocks.wl` checks the basic Dirac conventions and defines the current vertices. The script `scripts/step2_hard_photon_numerators.wl` computes numerator-level hard-photon traces in the explicit photon-vertex representation. These traces are stored in `outputs/step2_hard_photon_numerators.txt`.

The script `scripts/step2_e1_project_hard_numerators.wl` applies the E1 projector

$$\mathcal{P}_{E1}^{\mu\nu} = \frac{p^{\nu}q^{\mu} - (p \cdot q)g^{\mu\nu}}{2(p \cdot q)^2}, \quad (30)$$

which satisfies $\mathcal{P}_{E1}^{\mu\nu}[p_{\nu}q_{\mu} - (p \cdot q)g_{\mu\nu}] = 1$ for $q^2 = 0$. The projected hard numerators are stored in `outputs/step2_e1_projected_hard_numerators.txt`.

The script `scripts/step2_soft_2p_fierz_traces.wl` computes the trace factors multiplying the two-particle photon DA matrix elements after the Fierz rearrangement. The output `outputs/step2_soft_2p_fierz_traces.txt` shows which bilinears feed the axial current and which feed the tensor current.

7 Stage 1 Axial-Current Numerical Baseline

Before deriving the tensor-current sum rule, we implemented the known single-axial-current result for $D_{s1}(2460) \rightarrow D_s \gamma$ as a benchmark. This is the coupling called g_1 by Colangelo, De Fazio, and Ozpineci. In our notation this is the axial-current form factor G_A .

The Stage 1 version contains

$$G_A^{\text{Stage 1}} = G_A^{\text{pert}} + G_A^{\chi\phi\gamma} + G_A^{\mathbb{A},\bar{H}\gamma} + G_A^{f_{3\gamma}\Psi^v}, \quad (31)$$

where the three-particle DA integral usually denoted by $\mathcal{F}_1$ is omitted until Stage 2.

The numerical script is

`scripts/stage1_axial_g1_baseline.py.`

It writes the full Borel/threshold grid to

`outputs/stage1_axial_g1_baseline.csv.`

7.1 Magnetic-susceptibility sign convention

The Colangelo paper prints $\chi = -(3.15 \pm 0.30) \text{ GeV}^{-2}$ in its matrix-element convention. In this project we store the BBK magnitude as

$$\chi(1 \text{ GeV}) = +3.15 \pm 0.30 \text{ GeV}^{-2}. \quad (32)$$

Therefore the sign of the $\chi\phi_\gamma$ term in the implemented sum rule is flipped relative to the literal printed formula. This is a convention translation, not a physical change. A diagnostic run that inserts positive χ into the printed negative- χ formula gives a much larger width and is kept only as a warning check.

7.2 Preliminary Stage 1 result

Using the project inputs, $u_0 = 1/2$, $3 \leq M^2 \leq 6 \text{ GeV}^2$, and $s_0 = (2.50, 2.55, 2.60)^2 \text{ GeV}^2$, the axial-current Stage 1 grid gives

$$-0.399 \leq G_A^{\text{Stage } 1} \leq -0.314 \text{ GeV}^{-1}. \quad (33)$$

With

$$\Gamma = \frac{\alpha_{\text{em}}}{3} G^2 |\vec{q}_\gamma|^3, \quad (34)$$

this corresponds to

$$20.8 \text{ keV} \leq \Gamma_A^{\text{Stage } 1} [D_{s1}(2460) \rightarrow D_s \gamma] \leq 33.5 \text{ keV}. \quad (35)$$

This is not yet the physical mixed-state result. It is the axial-current baseline that will be combined with the tensor-current form factor after the J_B sum rule is derived.

8 First Tensor-Current Step: Leading-Twist Soft Contribution

The tensor-current calculation starts from the trace-level identity produced by `scripts/step3_soft_2p_e1_pro`. For the two-particle tensor photon DA matrix element, the E1 projections are

$$\mathcal{P}_{E1} [\text{Tr}(J_A S_Q i\gamma_5 \sigma_{\rho\lambda})(\varepsilon^\rho q^\lambda - \varepsilon^\lambda q^\rho)] = 8 \frac{k \cdot q}{p \cdot q}, \quad (36)$$

$$\mathcal{P}_{E1} [\text{Tr}(J_B S_Q i\gamma_5 \sigma_{\rho\lambda})(\varepsilon^\rho q^\lambda - \varepsilon^\lambda q^\rho)] = 8 \frac{m_Q}{m_Q + m_s}. \quad (37)$$

After the two-particle DA Fourier integral and double Borel transform, $k = p + \bar{u}q$, hence $k \cdot q = p \cdot q$ for $q^2 = 0$. Therefore, before the hadronic decay-constant normalization, the leading-twist tensor-current OPE piece is

$$\Pi_B^{\chi\phi_\gamma} = \frac{m_Q}{m_Q + m_s} \Pi_A^{\chi\phi_\gamma}. \quad (38)$$

The exploratory script `scripts/stage1_tensor_gb_leading_twist.py` uses this relation to estimate only the $\chi\phi_\gamma$ part of G_B . This is not the final tensor-current sum rule. It omits the tensor-current hard contribution, the vector DA contribution, and the Stage 2 three-particle DA terms.

Using the project inputs, the partial leading-twist estimate gives

$$G_B^{\chi\phi_\gamma} \simeq +0.205 \text{ to } +0.273 \text{ GeV}^{-1}, \quad \text{if } f_B = f_T, \quad (39)$$

$$G_B^{\chi\phi_\gamma} \simeq +0.114 \text{ to } +0.151 \text{ GeV}^{-1}, \quad \text{if } f_B = f_T \frac{m_{D_{s1}}}{m_c + m_s}. \quad (40)$$

The sizeable sign difference relative to G_A means that the physical mixed amplitudes can be sensitive to the tensor-current normalization. This is why the next step is to finish the full G_B sum rule before quoting final mixed widths.

9 Tensor-Current Hard Term: Triangle Reduction

The hard tensor-current numerator is more complicated than the axial one because the J_B current introduces an extra factor of P^α and therefore higher powers of the loop momentum after E1 projection. The script `scripts/step3_tensor_hard_triangle_reduction.wl` rewrites scalar products such as k^2 , $k \cdot p$, and $k \cdot q$ in terms of inverse propagator denominators. For the heavy-line photon emission we define

$$d_{H1} = (k + q)^2 - m_Q^2, \quad (41)$$

$$d_{H2} = k^2 - m_Q^2, \quad (42)$$

$$d_{H3} = (k - p)^2 - m_s^2, \quad (43)$$

and similarly for the strange-line photon emission. Setting $d_{Hi} = 0$ isolates the genuine three-denominator triangle core. For the tensor current this gives

$$N_{B,H}^{\text{core}} = \frac{-8im_Q^2 + 4im_Qm_s}{m_Q + m_s} - \frac{4im_Q^2p^2}{(m_Q + m_s)(p \cdot q)}, \quad (44)$$

$$N_{B,s}^{\text{core}} = \frac{-4im_Qm_s}{m_Q + m_s} - \frac{4im_s^2p^2}{(m_Q + m_s)(p \cdot q)}. \quad (45)$$

The same reduction was also performed for the axial current in `scripts/step3_axial_hard_triangle_reduction.wl`. This is a useful calibration because the axial hard term has a known published spectral density.

9.1 Why denominator-cancellation terms are separated

Terms proportional to one or more inverse denominators cancel one or more propagators. After such a cancellation the loop integral contains at most two propagators. These reduced integrals depend on at most one of the two external virtualities, p^2 or $(p + q)^2$, and therefore do not generate the double-pole structure selected by the double Borel transform. They are recorded in the output files for auditability, but the double-pole sum rule is built from the three-denominator triangle core.

This observation narrows the remaining hard-tensor task: derive the double spectral density associated with the tensor-current triangle core and match it to the same E1 invariant amplitude used for the axial-current benchmark.

10 First Hard-Tensor Spectral Density Candidate

The next step is to translate the triangle core into a perturbative spectral density. We do this by calibrating against the known axial-current result. For one photon-emitting line, let m_i be the emitting-quark mass and m_j the spectator mass. The axial triangle core has the schematic form

$$N_i^{A,\text{core}} = -4i \left[a_i + \frac{b_i}{p \cdot q} \right]. \quad (46)$$

Comparing with the published axial spectral density fixes the replacement

$$-4i a_i \longrightarrow 2a_i \log \frac{s - m_j^2 + m_i^2 - \lambda^{1/2}(s, m_j^2, m_i^2)}{s - m_j^2 + m_i^2 + \lambda^{1/2}(s, m_j^2, m_i^2)}, \quad (47)$$

$$-4i \frac{b_i}{p \cdot q} \longrightarrow \frac{b_i}{m_i^2} \frac{m_j^2 - m_i^2 - s}{s^2} \lambda^{1/2}(s, m_j^2, m_i^2). \quad (48)$$

The overall factor is $-3/(8\pi^2)$, and the total axial density is assembled as the strange-emission contribution minus the heavy-emission contribution, with the corresponding charges. The script

scripts/stage1_tensor_gb_hard_candidate.py

first applies these rules back to the axial current and reproduces Colangelo's $\rho^P(s)$ with a maximum numerical difference 1.4×10^{-17} on the test grid.

For the tensor current the core contains $p^2/(p \cdot q)$:

$$N_{B,H}^{\text{core}} = -4i \left[\frac{2m_Q^2 - m_Q m_s}{m_Q + m_s} + \frac{m_Q^2 p^2}{(m_Q + m_s)(p \cdot q)} \right], \quad (49)$$

$$N_{B,s}^{\text{core}} = -4i \left[\frac{m_Q m_s}{m_Q + m_s} + \frac{m_s^2 p^2}{(m_Q + m_s)(p \cdot q)} \right]. \quad (50)$$

In the first candidate implementation we use the diagonal double-dispersion identification $p^2 \rightarrow s$ inside this numerator. This is the remaining point that should be independently checked from the double spectral representation.

Combining this hard candidate with the tensor-DA two-particle soft contribution gives, over the same M^2 and s_0 windows,

$$\begin{aligned} G_B^{\text{hard}} &\simeq -0.831 \text{ to } -0.673 \text{ GeV}^{-1}, \quad G_B^{\text{Stage 1}} \simeq -0.510 \text{ to } -0.440 \text{ GeV}^{-1}, \quad (f_B = f_T), \quad (51) \\ G_B^{\text{hard}} &\simeq -0.460 \text{ to } -0.373 \text{ GeV}^{-1}, \quad G_B^{\text{Stage 1}} \simeq -0.283 \text{ to } -0.244 \text{ GeV}^{-1}, \quad \left(f_B = f_T \frac{m_{D_{s1}}}{m_c + m_s} \right). \quad (52) \end{aligned}$$

At $\theta = 35.3^\circ$, the corresponding candidate mixed widths are

$$\Gamma[D_{s1}(2460) \rightarrow D_s \gamma] \simeq 62.5 \text{ to } 87.9 \text{ keV}, \quad \Gamma[D_{s1}(2536) \rightarrow D_s \gamma] \simeq 0.00 \text{ to } 0.30 \text{ keV}, \quad (f_B = f_T), \quad (53)$$

$$\Gamma[D_{s1}(2460) \rightarrow D_s \gamma] \simeq 30.9 \text{ to } 44.7 \text{ keV}, \quad \Gamma[D_{s1}(2536) \rightarrow D_s \gamma] \simeq 4.0 \text{ to } 8.2 \text{ keV}, \quad \left(f_B = f_T \frac{m_{D_{s1}}}{m_c + m_s} \right). \quad (54)$$

These numbers should be read as a controlled Stage 1 candidate, not yet as the final result, because Stage 2 and the independent double-spectral check of the $p^2/(p \cdot q)$ term remain.